CS6-DAMPER MASS

Brief: Coming into the 2010 season, some materials have been made illegal by FIA regulations. This includes our current material for the camshaft pendulum dampers. The material must now be below a density of 18,100 kg/m3. Packaging of the carousel is restricted to18mm diametrically and you may fit as many pendulum dampers as you like per carousel. The aim of this project is to find the solution that will produce the least wear on the damper masses or carousel while matching the damping torque and vibration order. The camshaft carousel and washer to assemble the system are made of EN9 steel.

Their solution:

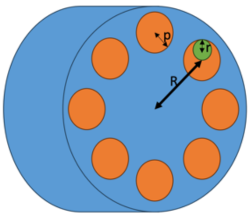
Damper Mass: Tungsten Alloy – Meets density requirement, High wear and thermal resistance

Carousel: EN9 Steel – Good hardness and machinability, cost-effective and well matched to damper material

What is a damper mass?

Dampers, also known as shock absorbers, control the movement of the springs and prevent the car from bouncing or oscillating. In Formula 1, dampers are usually gas-filled and adjustable, allowing the driver to calibrate the suspension for distinct track conditions. Moreover, they convert the suspension's kinetic energy into heat, then depleted into the atmosphere. Such action helps keep the system cool and prevent overheating, causing the car to lose grip and stability.-help maintain tyre contact with the road, preventing excessive bouncing-provides stability to the car.

How does it work?



- Uses centrifugal force and inertia to resist engine oscillations.
- Mass swings within a bore, causing reactive damping torque.
- Mounted inside a carousel, retained with washers; the mass moves relative to rotation.
- Damping torque is tuned by changing the mass, radius, and number of dampers.

Two types were highlighted for this case study

Rotational damper	Pendulum damper
• Oppose vibrations in a torsional system to reduce the fatigue on components. • The less the component moves back and forth the longer it will last and the smoother the movement in the power unit will be.	• Oppose this movement with a high-density weight • Weight swings in the opposite direction to motion. E.g. as the carousel below (blue) rotates clockwise the damper mass (green) will rotate anti-clockwise in the bore (orange).

Physical Assembly

Each damper mass is placed inside a bore or cavity within a larger rotating structure such as a camshaft carousel. The mass is constrained by two end washers, one on each side, which keep it axially fixed while still allowing it to rotate slightly within the bore. The washers and the carousel are made of the same material, typically EN9 steel, known for its balance of strength and hardness. This ensures the assembly can withstand repeated loading without excessive wear. Lubrication is also critical, as the operating environment involves high temperatures (up to 140°C) and oil exposure.

How should you choose the materials for the damper mass and the carousel that it sits in?

Requirements: Brief 5 • Less than a density of 18,100 kg/m3. • Least wear on the damper masses or carousel • Match the damping torque and vibration order. • Rolling face interaction with EN9 steel. (Hardness = 150-350HV) • Operating in a power unit – exposed to temperatures up to 140 degC, oil, vibration within the carousel

Objectives	Constraints	Free variables
Wear Resistance – high ((Hardness * Contact Area) / (Sliding Velocity * Load)) Hardness – high Density – high Yield Strength – high	Density < 18,100 kg/m3 Material types i.e. must be a solid Hardness around EN9 steel (100 < HV < 400) Max Temperature > 140 degC Corrosion resistant to oil	cost

Material	Density (kg/m3)	Hardness (HV)	Yield Strength (MPa)	Cost (£ per kg)
Tungsten Alloys	17000-18500	300-400	500-1000	75-230
Tungsten Carbide	14500	1600-2500	300-350	200-300
Nickel Alloys	8000-9000	100-400	150-170	40-75
Steel	7000-8000	100-800	200-500	100-200

What materials should you use and explain why?

Taking into account the design objectives and constraints, and according to ashby charts of the effective modulus against Hardness, and density against strength,these materials came out as top options.

Chosen material :Tungsten Alloys

- **Density** is very close to the 18,100 kg/m^3 limit → lets you achieve high torque in compact packaging.
- **Hardness (300–400 HV)** is **well-matched to EN9 steel** (150–350 HV), avoiding excessive wear.
- **High yield strength (500–1000 MPa)** ensures resistance to deformation under vibration/load.
- **Wear resistance is solid** without being so hard that it damages the carousel or washers.
- **Available as a machinable alloy**, unlike brittle tungsten carbide.

Others:

- **Tungsten Carbide**: *Too hard* → would damage EN9 surfaces and is expensive.
- **Nickel Alloys**: Easier to machine, but low hardness and yield strength reduce performance under load.
- **Steel**: Lowest density → would require larger size/more dampers to achieve same torque

How do the vibration and torque calculations work?

Variable	Damper torque	Vibration order
Mass-higher	Higher	-
Bore radius-higher	Higher	Higher
Roller radius -higher	Lower	Lower (order of 2)
Number of dampers-higher	Lower	

Damper Mass is a simple cylinder

▪ Radius = designable

▪ Length = 15mm

- Damper Torque:
  - $\frac{m \times \omega_{engine} \times R \times 0.5 \times (R+e-\frac{25}{r})}{1+(\frac{25}{r})^2} \times n_{dampers}$
  - **Your damper torque should equal as close to 19.67Nm @ 11000rpm**
- Vibration Order:
  - $(\frac{R}{e \times (1+(\frac{37.5}{r^2}))})^{1/2}$
  - **The vibration order of this system is 4**

- The **torque** generated by the pendulum mass depends on **how far out (R + e)** and **how heavy (m)** the mass is.
- The **nonlinear denominator** 1+(25r)21 + \left(\frac{25}{r}\right)^21+(r25)2 accounts for dynamic interaction, reducing torque as roller radius increases.
- More **dampers** multiply the torque output linearly.

- Vibration order tells **how many times the damper completes a cycle per engine revolution**.

- Higher **R** or lower **e** means **faster angular response**.
- As **roller radius (r)** increases, the denominator increases (since 1r2\frac{1}{r^2}r21), **reducing vibration order**
- A **4th order** means the damper is tuned to absorb torsional vibrations that occur **four times per engine rev** — common in multi-cylinder engines where harmonics align with firing pulses.
- **vibration order** is something you want to **match to a specific harmonic of the engine's torsional vibration**

Torque calculations determine how much force the dampers can apply to resist twisting forces in the camshaft. It depends on damper mass, carousel radius, and rotational speed.
Vibration calculations ensure that the damper's frequency response aligns with the engine's vibration characteristics. This is vital for targeting and cancelling specific harmonics, like the 4th order torsional mode.

- 1. Generates sufficient torque (≈19.67 Nm)
- 2. Is tuned to the correct vibration order (4)

How should we surface treat the damper mass to improve lifetime?

To enhance the performance and longevity of rotational and pendulum dampers, careful material selection and surface engineering are critical. Components subject to rolling or sliding contact—such as damper masses and carousel interfaces—should be made from materials with high inherent wear resistance, such as hardened steels or advanced alloys.
To further increase durability and reduce friction, surface treatments and coatings can be employed. Options like chrome plating, anodizing, or advanced Physical or Chemical Vapor Deposition (PVD/CVD) coatings—especially Diamond-Like Carbon (DLC)—offer excellent wear resistance and reduce surface degradation over time. Additionally, appropriate heat treatments can improve the hardness and fatigue resistance of base materials.
Proper lubrication is essential to minimize direct metal-to-metal contact, thereby reducing friction and slowing wear progression. Lubricants must be compatible with both the material and the operating temperature range of the damper. Equally important is
precision in mechanical design: ensuring sufficient clearance between interacting components prevents binding or localized wear. The surface finish of critical parts greatly influences frictional behavior; therefore, high-precision machining methods should be
selected for the carousel bores and damper masses to ensure smooth operation and consistent vibration damping performance.

How do you design to counter fretting / abrasion / wear in practice?

Material Selection

• Use materials with higher wear resistance for the rolling/sliding components.
• Applying coatings such as chrome plating or specialized wear-resistant coatings can help reduce friction and protect surfaces from wear.
For example, heat treatments, anodising or PVD/CVD (DLC - Diamond-Like Carbon) coatings can improve durability.

Lubrication

• Ensure that proper lubrication is applied to the moving components. This helps reduce wear by minimizing direct contact between moving parts.

Friction Control

• Ensure that the damper is designed so there is enough clearance between components. Surface finish of materials will make a huge difference, therefore machining methods should be considered for both carousel and damper mass.

How might you make an individual damper mass?

1. Manufacturing the Damper Mass (Tungsten Alloy)

Step-by-Step:

1. **Material Selection & Procurement**:
   - Obtain high-density **tungsten alloy rod/bar stock** (e.g., W-Ni-Fe alloy).
   - Ensure certification of properties (density, hardness, yield strength).
2. **Precision Turning**:

- Use **CNC lathe** to machine the damper into a **cylindrical shape**:
  - Length = 15 mm
  - Radius = optimized via damping torque equation
- Maintain **tight tolerances** (e.g., ±0.01 mm) for dynamic balance.

3. **Surface Finishing**:
   - **Grinding and polishing** to reduce surface roughness and wear.
   - Optional **surface coatings** (DLC, PVD) to reduce friction against steel.
4. **Quality Control**:
   - **Mass check** (to ensure proper damping torque).
   - **Hardness test** (Vickers or Rockwell).
   - Dimensional check (CMM or micrometer).

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2. Manufacturing the Carousel (EN9 Steel)

Step-by-Step:

1. **Material Preparation**:
   - Cut EN9 steel billet or bar to rough size.
2. **CNC Machining**:
   - **Mill or turn** the carousel to final external geometry.
   - **Drill/ream multiple bores** to house damper masses.
   - **Include slots or shoulders** to accommodate washers/retainers.
3. **Heat Treatment** *(optional)*:
   - Apply **normalizing or hardening + tempering** to reach desired mechanical properties (150–350 HV, good toughness).
4. **Surface Finish & Tolerances**:
   - Ensure smooth bore surfaces for damper swing.
   - Use **precision boring** or **honing** for critical bore dimensions.
5. **Washer & Bolt Assembly**:
   - Machine **washers** from EN9 or stainless steel.
   - Insert **retaining bolts** to trap dampers securely while allowing swing.
6. **Dynamic Balancing**:
   - Final **balancing test** to prevent high-speed vibration during operation.

Assembly Notes

- Lubricate bores with engine-grade oil.
- Ensure **damper masses rotate freely** in the bores without binding.
- Verify the **clearance** and **coaxial alignment** of the bores with carousel center.